pacemaker

Software Requirements Specification

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Contents

Introduction

This document specifies the software requirements for the **pacemaker** library, a Windows COM automation wrapper that provides a safe, idiomatic C++ interface to the INCA calibration and measurement tool.

The library is organised around a thin proxy layer: each COM interface exported by INCA is wrapped in a dedicated C++ class that owns the underlying COM pointer via a `unique_com_ptr` RAII handle. A `Session` object composes the proxies into a higher-level experiment workflow.

The requirements below are grouped by header file / component.

REQ-001 – `unique_com_ptr` (RAII COM Pointer)

Title:

RAII ownership wrapper for COM objects `[implemented]`

Intent:

The user needs a way to hold COM object pointers whose lifetime is automatically managed. When the wrapper goes out of scope the COM object's `Release()` method must be called exactly once, preventing resource leaks without requiring explicit cleanup code at every call site.

Acceptance criteria:

> **AC-001-1**
>
> `unique_com_ptr<T>` shall call `T::Release()` on destruction when the stored pointer is non-null.

> **AC-001-2**
>
> `unique_com_ptr<T>` shall *not* call `Release()` when the stored pointer is `nullptr`.

> **AC-001-3**
>
> The type shall be move-constructible and move-assignable; move shall transfer ownership without calling `Release()` on the moved-from value.

> **AC-001-4**
>
> The type shall be non-copyable.

> **AC-001-5**
>
> `get()` shall return the raw pointer without affecting ownership.

> **AC-001-6**
>
> The type shall be compatible with `std::unique_ptr` semantics (i.e. implemented as an alias to `std::unique_ptr<T, ComObjDeleter<T»`).

Non-functional constraints:

- Platform: Windows only (COM is a Windows technology).
- Zero overhead: the deleter is a stateless functor; no heap allocation per wrapper.

Testing:

See test document, section *TC-001*.

REQ-002 – `COMProxy<T>` (Base Proxy Template)

Title:

Generic non-copyable base class for COM dispatch proxies `[implemented]`

Intent:

The user wants a single, reusable base that stores the typed COM pointer and enforces ownership rules, so that concrete proxy classes do not repeat boilerplate and cannot accidentally be copied (which would silently duplicate COM reference counts).

Acceptance criteria:

> **AC-002-1**
>
> `COMProxy<T>` shall be copy-constructible and copy-assignable only when explicitly deleted (not available to callers).

> **AC-002-2**
>
> `COMProxy<T>` shall be move-constructible and move-assignable.

> **AC-002-3**
>
> The constructor shall accept a `unique_com_ptr<T>` by value (taking ownership via move).

> **AC-002-4**
>
> The protected member `p_subject` shall be accessible to derived classes.

> **AC-002-5**
>
> The destructor shall be non-virtual and defaulted; derived classes are responsible for their own destructors.

Testing:

See test document, section *TC-002*.

REQ-003 – `IncaProxy`

Title:

Proxy for the top-level INCA COM object `[implemented]`

Intent:

The user needs a C++ object that connects to a running INCA instance and retrieves the currently open experiment and experiment-view COM objects. On destruction the proxy must politely disconnect from the tool so that INCA is not left in a stale connected state.

Acceptance criteria:

> **AC-003-1**
>
> The constructor shall narrow the supplied `IDispatch` to `Inca_Dispatch` via `query_interface`; it shall throw if the narrowing fails.

> **AC-003-2**
>
> The destructor shall call `DisconnectFromTool()` if the internal pointer is non-null.

> **AC-003-3**
>
> `GetOpenedExperiment()` shall return a `unique_com_ptr<IDispatch>` owning the experiment object.

> **AC-003-4**
>
> `GetOpenedExperimentView()` shall return a `unique_com_ptr<IDispatch>` owning the experiment-view object.

> **AC-003-5**
>
> The class shall be move-only (copy deleted).

Testing:

See test document, section *TC-003*.

REQ-004 – `IncaOnlineExperimentProxy`

Title:

Proxy for the INCA online experiment COM object `[implemented]`

Intent:

The user needs to start and stop measurement recordings, enumerate connected ECU devices, and retrieve calibration data items from a specific device, all through a safe C++ API that hides COM SAFEARRAY marshalling and BSTR string handling.

Acceptance criteria:

> **AC-004-1**
>
> The constructor shall narrow the supplied `IDispatch` to `IncaOnlineExperiment_Dispatch` via `query_interface`.

> **AC-004-2**
>
> `GetAllDevices()` shall return a `std::vector` of `unique_com_ptr<IDispatch>`, one element per device reported by INCA.

> **AC-004-3**
>
> `GetAllDevices()` shall throw `std::runtime_error` if the returned SAFEARRAY is not of type `VT_ARRAY | VT_VARIANT` or if any element is not `VT_DISPATCH`.

> **AC-004-4**
>
> `StartRecording()` shall delegate to the underlying COM method without additional side-effects.

> **AC-004-5**
>
> `StopRecordingAndSave()` shall delegate to the underlying COM method.

> **AC-004-6**
>
> `StopMeasurement()` shall delegate to the underlying COM method.

> **AC-004-7**
>
> `GetCalibrationValueInDevice(name, device)` shall return a `unique_com_ptr<IDispatch>` owning the calibration object, and shall throw `std::runtime_error` if the returned pointer is null.

> **AC-004-8**
>
> The class shall be move-only.

Testing:

See test document, section *TC-004*.

REQ-005 – `IncaExperimentViewProxy`

Title:

Proxy for the INCA experiment-view COM object `[implemented]`

Intent:

The user needs to programmatically open a measurement view for a specific calibration data item inside INCA, so that data can be observed in the INCA GUI during an automated experiment run.

Acceptance criteria:

> **AC-005-1**
>
> The constructor shall narrow the supplied `IDispatch` to `IncaExperimentView_Dispatch` via `query_interface`.

> **AC-005-2**
>
> `OpenViewForExperimentDataItem(dataitem)` shall forward the raw `IDispatch*` pointer (without transferring ownership) to the underlying COM method.

> **AC-005-3**
>
> The class shall be move-only.

Testing:

See test document, section *TC-005*.

REQ-006 – `ExperimentDeviceProxy`

Title:

Proxy for an INCA experiment device COM object `[implemented]`

Intent:

The user needs to identify a connected ECU device by name and pass its raw COM pointer to other INCA methods that require an `ExperimentDevice_Dispatch*` argument, all without managing COM reference counts manually.

Acceptance criteria:

> **AC-006-1**
>
> The constructor shall narrow the supplied `IDispatch` to `ExperimentDevice_Dispatch` via `query_interface`.

> **AC-006-2**
>
> `GetName()` shall return a `std::wstring` with the device name obtained from the underlying BSTR.

> **AC-006-3**
>
> `get()` shall return a non-owning raw pointer (`ExperimentDevice_Dispatch*`) suitable for use as a COM method argument; it shall be marked `noexcept`.

> **AC-006-4**
>
> The class shall be move-only.

Testing:

See test document, section *TC-006*.

REQ-007 – `CalibrationScalarDataProxy`

Title:

Proxy for an INCA calibration scalar data COM object `[implemented]`

Intent:

The user needs to set a scalar calibration parameter to a specific numeric value during an experiment, and to restore it to its reference-page value when the experiment ends, through a simple C++ interface that hides the underlying COM dispatch calls.

Acceptance criteria:

> **AC-007-1**
>
> The constructor shall narrow the supplied `IDispatch` to `CalibrationScalarData_Dispatch` via `query_interface`.

> **AC-007-2**
>
> `SetImplValue(value)` shall forward the `double` to the underlying COM `SetImplValue` method.

> **AC-007-3**
>
> `ResetValueToRP()` shall call the underlying COM `ResetValueToRP` method.

> **AC-007-4**
>
> The class shall be move-only (move constructor and move assignment defaulted; copy deleted implicitly via base).

Testing:

See test document, section *TC-007*.

REQ-008 – `Experiment`

Title:

High-level experiment workflow manager `[implemented]`

Intent:

The user needs a single, coherent object to manage the full lifecycle of an INCA measurement run: start and stop recording, register named calibration parameters, update their values during the run, and reset all parameters to their reference-page values when the experiment concludes. The object shall enforce correct usage order and provide clear error messages when the API is misused.

Acceptance criteria:

> **AC-008-1**
>
> `start_recording()` shall delegate to `IncaOnlineExperimentProxy::StartRecording()`.

> **AC-008-2**
>
> `stop_recording(filename)` shall: wait at least 40 ms (flush delay), call `StopRecordingAndSave()`, call `StopMeasurement()`, and then call `reset()`.

> **AC-008-3**
>
> `add_param(name)` shall retrieve the calibration scalar object from INCA and store it internally. Calling `add_param` a second time with the same name shall be a no-op.

> **AC-008-4**
>
> `set_param(name, value)` shall set the calibration value. If `name` was not previously registered via `add_param`, it shall throw `std::out_of_range` with a descriptive message.

> **AC-008-5**
>
> `reset()` shall call `ResetValueToRP()` on every registered proxy, then clear the internal map and vector.

> **AC-008-6**
>
> The class shall be move-constructible; copy shall be deleted.

> **AC-008-7**
>
> The flush delay constant `k_flush_delay` shall be 40 ms.

Non-functional constraints:

- The 40 ms flush delay is required to ensure INCA has time to write all buffered measurement data before the recording is stopped.

Testing:

See test document, section *TC-008*.

REQ-009 – `Session`

Title:

Top-level session factory and lifetime manager `[implemented]`

Intent:

The user needs a single entry point that establishes a COM connection to a running INCA instance, validates that the environment is ready (experiment open, device online), and returns a fully-initialised object through which the experiment can be controlled. The session shall be non-copyable and non-movable to prevent accidental duplication of COM connections.

Acceptance criteria:

> **AC-009-1**
>
> `Session::connect()` shall call `CoCreateInstance` with the `CLSID_Inca` class identifier and `CLSCTX_LOCAL_SERVER`.

> **AC-009-2**
>
> If `CoCreateInstance` fails, `connect()` shall throw `std::runtime_error` with a message indicating that INCA may not be installed.

> **AC-009-3**
>
> `connect()` shall retrieve the open experiment and experiment-view objects via `IncaProxy`.

> **AC-009-4**
>
> `connect()` shall enumerate devices; if none are found it shall throw `std::runtime_error` with a message indicating no device is configured or online.

> **AC-009-5**
>
> On success, `connect()` shall return a `Session` owning an `IncaProxy` and an `Experiment` built from the first available device.

> **AC-009-6**
>
> `experiment()` shall return a non-const reference to the owned `Experiment`.

> **AC-009-7**
>
> The class shall be non-copyable and non-movable.

Non-functional constraints:

- COM must be initialised by the caller before invoking `connect()` (e.g. via `CoInitializeEx`).

- Platform: Windows only.

Notes:

The `Session` destructor is explicitly defined (even if empty) to ensure that the `IncaProxy` destructor (which calls `DisconnectFromTool`) is invoked at a predictable point during session teardown.

Testing:

See test document, section *TC-009*.